The Kingman *et al.* 2021 stickleback dataset

Acquisition, packaging, and overlap with the 3sp LD-aware outlier regions

LDscnR-paper · `kingman2021/`

August 2026

1 Purpose

This module acquires and packages the data of Roberts Kingman *et al.* 2021, *Predicting future from past: the genomic basis of recurrent and rapid stickleback evolution*, Sci Adv 7:eabg5285, as the **clear-peak** counterpart to the **saturated** 3sp dataset already analysed in `module_sticklebacks_LDscnR/`.

3sp is the case where the whole genome carries real outlier regions and there is no neutral floor. What the C-score machinery still needs is a dataset with a genuinely quiet background against which the detectability gate and the negative-control-floor τ_C calibration can be exercised. Kingman 2021 supplies exactly that: recurrent marine→freshwater adaptive loci standing sharply above a quiet background, with a published, independently derived peak set to score against.

2 Acquisition

2.1 The processed VCF exists, so no variant calling was needed

The paper's Data Availability statement offers only raw SRA reads (PRJNA247503: 1758 runs, 206 individuals, 0.7 TB) and a track hub at `sbwdev.stanford.edu`, which is dead — no ICMP reply and TCP timeouts on both 80 and 443, from this network and through an off-network proxy.

However, the hub description page at `https://web.stanford.edu/group/kingsley/hubDescription.html` notes that the entire assembly hub is mirrored on **FigShare project 162634**. That mirror contains `227_genomes.final.filtered.vcf.gz` (15.3 GB + `.tbi`): a GATK-called, VQSR-filtered joint VCF of all 227 genomes, already in gasAcu1-4 coordinates. The whole alignment-and-calling path was therefore superseded and never run.

2.2 Sources that could not be used

Dryad (`10.5061/dryad.pvmcvdnjm`, `10.5061/dryad.547d7wm6t`) now returns {`"error":"Unauthorized, must have current bearer token"`} from its API, and its web download route sits behind an Anubis anti-bot challenge, which was not circumvented. The reference genome was obtained from FigShare instead. Still unobtained are the 363-SNP genotyping array and the published liftOver chains; neither is needed — the array is a validation panel rather than discovery data, and the chains were reconstructed (§4). **science.org** blocks scripted access, so Table S2 was retrieved through a browser session.

3 The dataset

3.1 Cohorts are one genome per population

This is the design fact that governs everything downstream. The 227 genomes are *not* population samples; for the EcoPeak analyses they are one individual per population, and Table S2 encodes the cohort membership, with the flag value being the ecotype (1 = marine, 0 = freshwater) and blank meaning "not in this cohort".

Cohort	Geography	n	M / F	Published peak set
c155_global	global	84	28 / 56	Global specific: 39 peaks, 3.74 Mb (0.8% of genome)
c150_pacNW	N.E. Pacific	68	11 / 57	Pacific specific: 209 peaks, 27.4 Mb (5.9%)
c151_nEur	N. Europe	27	9 / 18	none published

The c155 membership reproduces the hub's stated "56 freshwater and 28 marine" exactly, and all 227 Table S2 `Sequence IDs` match the 227 VCF sample names by set equality. Every published peak-set region count and span was checked against the paper and matches (39/3.7 Mb, 209/27.4 Mb, 212/91.9 Mb).

3.2 Coverage and the consequences of it

Mean sequencing coverage is 5.5×. At `MAF` ≥ 0.05 and `F_MISSING` ≤ 0.20, roughly 12% of genotype calls are missing, and tightening the filter is not an escape: `F_MISSING` ≤ 0.05 retains only 3.8% of SNPs. The packaged `GTs` matrices are therefore **mean-imputed and numeric** rather than integer 0/1/2, because `gcta_grm()` and `emmax_setup()` have no NA handling. Per-SNP `f_missing` is carried in `map` so any downstream step can filter on it. The residual caveat is real and should be remembered when interpreting LD-based results: hard calls at 5.5× attenuate r^2 and hence `ld_w` relative to a high-coverage dataset. The VCF retains `PL`, so genotype-posterior dosages remain available if attenuation proves limiting.

3.3 Packaged bundles

Bundle	Individuals	SNPs	Ecotype
kingman2021_c155_global.rds	84	3,509,797	28 M / 56 F
kingman2021_c150_pacNW.rds	68	4,073,192	11 M / 57 F

Each is `list(GTs, map, eco, meta, cohort, source)` with `map` = `marker, Chr, chr_num, Pos, rec_rate, cM, af, f_missing`. `rec_rate` is the Rabbit Slough LDhelmet map in cM/Mb and `cM` its running integral; the map covers 96.4% of SNPs, the remainder falling in gaps and carrying `NA`. A matched `_truth.rds` carries the peak table and per-SNP peak membership, row-aligned with `map`.

3.4 The packaged data reproduces the published peaks

Naive $|\Delta \mathrm{AF}|$ between ecotypes, inside the specific EcoPeaks versus the rest of the genome:

Cohort	median in-peak	background	above background 99.9th pct	Enrichment
c155_global	0.179	0.058	29.8% vs 0.10%	**298×**
c150_pacNW	0.196	0.088	1.61% vs 0.10%	16×

And the background really is quiet, which is the property the dataset was acquired for: on chrI:1–3 Mb, which contains no EcoPeak, **0 of 24,551 SNPs** reach $p < 10^{-6}$ and 8 reach $p < 10^{-4}$ in an uncorrected ecotype regression, with $\lambda \approx 1.17$.

Use `c155_global` as the primary clear-peak dataset: balanced 28/56 design and a truth set covering only 0.8% of the genome. `c150_pacNW` is the secondary, offering more truth regions but only 11 marine individuals.

4 The coordinate bridge

3sp is in **gasAcu1** (`Chr1..Chr21`, arabic); everything Kingman is in **gasAcu1-4** (`chrI..chrXXI`, roman). Since the published chains were unobtainable, a chain was reconstructed from the hub's `bigChain` + `bigChain.link` tracks (1,095 chains, 19,851 blocks) and reversed with UCSC `chainSwap`.

It validates cleanly. Every peak set lifts at ≈100% with preserved span (c155 specific 3.743 → 3.743 Mb; c150 specific 27.44 → 27.18 Mb; c150 sensitive 91.90 → 90.69 Mb with 2 unmapped), chromosome assignment is preserved throughout, and the *Eda* EcoPeak at chrIV:12,795,310–12,892,195 lands at chrIV:12,782,916–12,879,801 — a 12 kb shift, as expected for two near-identical assemblies on that chromosome.

5 Overlap with the 3sp outlier regions

The comparison target is the 40 LFMM outlier regions from `module_sticklebacks_LDscnR` at $\tau_C = 0.05$, $l_{\min} = 10$, $\rho_{ld} = 0.60$, spanning 21.13 Mb (5.6% of the SNP-covered genome).

Kingman set	Peaks	Genome	Regions hitting $\geq$1 peak	Fold	p
Global specific	36	0.9%	4/40 (10%)	1.38×	0.33
Global sensitive	83	5.8%	13/40 (32%)	1.85×	**0.0065**
Pacific specific	193	6.6%	15/40 (38%)	1.47×	**0.029**
Pacific sensitive	189	21.7%	18/40 (45%)	1.13×	0.29

Null is a within-chromosome shuffle of the regions, $B = 2000$. By base pairs the picture is similar: 2.3% of LFMM span sits inside global-specific peaks against 0.9% expected (2.48×, $p = 0.08$). Reciprocally only 4 of 36 global-specific and 29 of 193 pacific-specific peaks are hit.

This is a weak correspondence, and the metrics disagree on significance. The explanation turns out to be geography rather than method failure (§7).

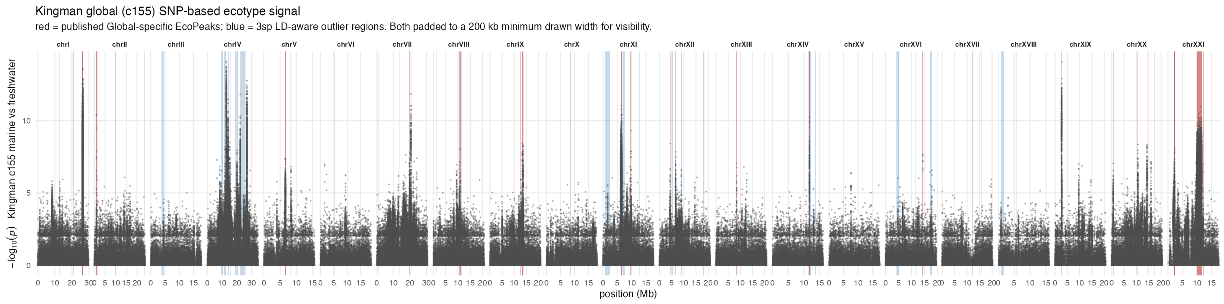

Figure 1: Kingman c155 SNP-based ecotype signal genome-wide. Red bands are the published Global-specific EcoPeaks; blue bands are the 3sp LD-aware outlier regions lifted into gasAcu1-4. Both are padded to a 200 kb minimum *drawn* width, since the median global-specific peak is 21 kb and would otherwise be sub-pixel; the data are unpadded. Their limited coincidence is the observation this report explains.

6 Re-deriving the Northern-European (c151) peak set

6.1 Method and validation

Kingman used two methods, intersected at 1% FDR for "specific" calls and unioned at 5% for "sensitive". Only the **SNP-based** test is specified precisely enough to reproduce: holding the genotype-class totals (n_0, n_1, n_2) and the marine group size m fixed, the marine ALT-allele count is scored against a multivariate-hypergeometric null,

$$P(a_0, a_1, a_2) = \frac{\binom{n_0}{a_0}\binom{n_1}{a_1}\binom{n_2}{a_2}}{\binom{N}{m}}, \qquad k = a_1 + 2a_2,$$

two-sided on $|k - \mathbb{E}[k]|$. The null depends only on (n_0, n_1, n_2, m), so it is computed once per distinct key and cached; that is what makes it tractable genome-wide. The companion window/CSS method is **not** implemented, since the published description does not give the CSS formula.

The implementation was validated against both published cohorts:

Cohort	Threshold	My peaks	Published *specific* recovered
`c155_global`	BH $q \leq 0.01$	286 / 15.5 Mb	**36/39 (92%)**
`c155_global`	BH $q \leq 10^{-4}$	46 / 3.83 Mb	34/39 (87%); 46/46 of mine hit a published peak
`c150_pacNW`	BH $q \leq 10^{-4}$	274 / 35.2 Mb	**195/209 (93%)**

At $q \leq 10^{-4}$ the c155 output (46 peaks, 3.83 Mb) is near-identical in size to the published specific set (39 peaks, 3.74 Mb). Because only one of the two methods is implemented, these sets are a sensitive superset rather than a reproduction of "specific".

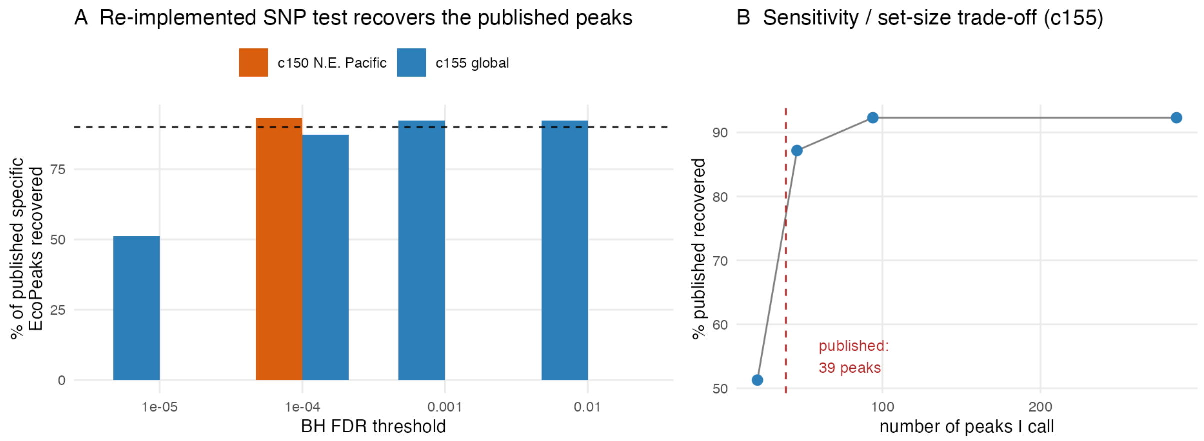

Figure 2: Validation of the re-implemented SNP-based test against the published peak sets.

6.2 c151 cannot reach genome-wide significance, and this is structural

c151 has 9 marine and 18 freshwater individuals, and after missingness the median marine count actually called is **7** (range 2–9). The exact test therefore bottoms out at $p = 3.2 \times 10^{-7}$, attained by exactly one SNP, whereas BH at $q \leq 0.01$ over 1.70 M tests would require ≥ 54 SNPs at that floor. **No SNP is significant at any FDR**, and no threshold choice repairs it.

This is very likely why the hub publishes EcoPeak BEDs for c150 and c155 only, even though Table S2 also defines c151, c153 and c154 cohorts. It is a property of the cohort, not of the re-implementation — which recovers 92–93% of the published peaks where power exists.

What remains usable is the *ranked* signal. A rank-based, explicitly **not FDR-controlled** set was emitted from the top 0.1% of p: 1,770 SNPs → **41 suggestive peaks, 1.31 Mb**.

7 The geographic gradient

Ranks sidestep the significance floor entirely. Testing whether the 3sp outlier regions (4.6–4.8% of tested SNPs) are enriched for low Kingman p, with a within-chromosome shuffle null:

Kingman cohort	vs 3sp geography	top 1%	p	top 0.1%	p
`c151` N. Europe	**matches**	**4.64×**	**0.0040**	1.07×	0.28
`c155` Global	partial	3.19×	0.024	6.30×	0.0080
`c150` N.E. Pacific	different basin	1.47×	0.22	0.71×	0.39

Independently, 11 of the 41 c151 suggestive peaks fall inside a 3sp outlier region against 2.80 expected: **3.93×,** $p = 0.0002$ (`R/11_c151_peak_overlap.R`).

3sp is an entirely northern-European Atlantic panel — lineage `NO`, sampled from the Baltic, North Sea, Norwegian Sea and White Sea, 90 freshwater and 27 marine. The published EcoPeaks it was first compared against are largely Pacific. Matching the ocean basin roughly triples the agreement. **The weak headline overlap is geography, not a failure of the LD-aware method.**

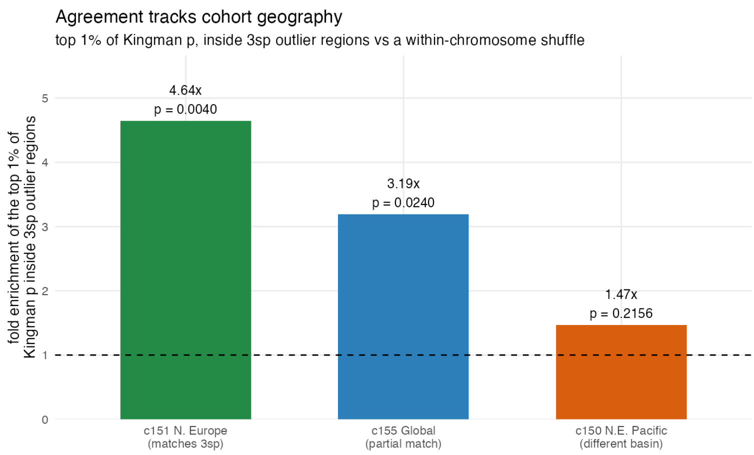

Figure 3: Agreement with the 3sp outlier regions tracks the geography of the Kingman cohort.

One honest qualification: the top-0.1% column disagrees with the top-1% column. c151's 1.70 M p-values take only 19,035 distinct values, so its extreme tail is shaped by which genotype configurations *can* attain a low p rather than by signal. The top-1% comparison is the trustworthy one.

8 Which regions actually overlap

8.1 The one locus everything agrees on

Chr1 21.499–21.929 Mb (gasAcu1) = chrI 26.1–26.5 Mb (gasAcu1-4) is recovered by four independent routes: the 3sp LFMM regions, the 3sp EMMAX regions (where it is the top region, 245 SNPs), the published Global-specific EcoPeak, and the c151 northern-European signal (its strongest suggestive peak). It is the Chr1 inversion.

8.2 *Eda*: recovered by EMMAX at $l_{\min}$=3, missed by LFMM entirely

The *Eda* gene sits at chrIV:12,812,613–12,822,840 (gasAcu1-4), i.e. 12,800,219–12,810,446 in gasAcu1, squarely inside a published EcoPeak. In the $l_{\min}$=10 LFMM set analysed above it

Region (gasAcu1)	Kingman peak	Overlap	Genes
Chr1 21.499–21.929	Global spec. (p=4.5e−15) *and* Pacific spec. (8.8e−12)	429 kb	*tns1, igfbp5, igfbp2a, stk11ip, cx44.2, inha, spega, ctdsp1*
Chr16 16.479–16.829	Pacific specific	350 kb	*fer1l6, prkag3b, ttll4, abcb6b, kalrna, ahr1b*
Chr4 22.732–23.052	Pacific specific	320 kb	*large1, mkrn1, caprin2, ipo8*
Chr4 20.985–21.278	Pacific specific	293 kb	*slc2a13, iapp, pyroxd1, etnk1*
Chr4 21.542–21.746	Pacific specific	203 kb	*mapk11, mapk12b, lrig3, slc16a7*
Chr4 11.368–11.535	Pacific specific	167 kb	*rack1, ctbp1, spon2a*
Chr4 14.428–14.563	Pacific specific	134 kb	*gria1a, med7, itk, cyfip2*
Chr11 6.247–6.370	Pacific specific	123 kb	*mrc2, rnd2, vat1, phb*
Chr12 11.874–11.986	Pacific specific	112 kb	*rhocb, wnt2b*
Chr4 14.313–14.373	Pacific specific	61 kb	*foxi3a, wnt8a*
Chr14 11.353–11.362	Global *and* Pacific spec.	9 kb	*cel*

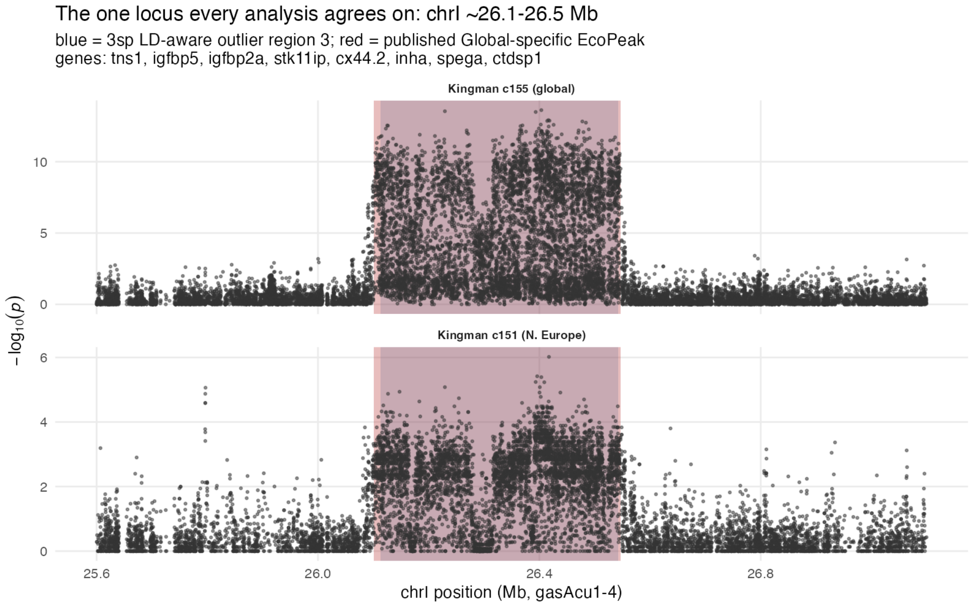

Figure 4: The shared chrI locus in the two Kingman cohorts. The 3sp outlier region (blue) and the published Global-specific EcoPeak (red) coincide so closely that the bands overlap. The signal block is sharply bounded in both cohorts, and present in c151 despite that cohort having no genome-wide-significant SNP anywhere.

falls in the *gap* between region 9 (ending 12.284 Mb) and region 10 (starting 14.034 Mb), and the four $l_{\min}$=10 EMMAX regions contain no Chr4 entry at all — which is what an earlier version of this report recorded as a flat absence. That was too strong. Relaxing the cluster-size floor resolves it, but only for one of the two methods:

The recovered EMMAX region overlaps the *eda* gene body by **1,371 bp**, and sits inside the Global-specific EcoPeak carrying the *most extreme SNP p-value in that entire set* (7.9×10^{-15}; the matching Pacific-specific peak is 2.2×10^{-12}).

Two things follow. First, this is a **method difference, not merely a threshold choice**: LFMM does not find *Eda* at either cluster-size floor, so lowering $l_{\min}$ alone does not rescue it. Second, it vindicates the claim in `module_sticklebacks_LDscnR/README.md` that the C-score resolves Chr4/*Eda* — the earlier report was simply reading it off the wrong region set. *Eda* is a 4-SNP cluster, so any analysis that demands ≥10 linked SNPs will discard the textbook marine–freshwater locus. That is a cautionary result about $l_{\min}$ in its own right.

Method	$l_{\min}$	*Eda* recovered?	Why
EMMAX	3	**yes**	region 7 = Chr4:12,809,075–12,811,617, **4 SNPs**
EMMAX	10	no	the same region has only 4 SNPs, below the floor
LFMM	3	no	Chr4 regions jump 12.284 → 14.034 Mb
LFMM	10	no	same gap

9 The $l_{\min}$=3 region sets

The analysis above uses the 40 LFMM $l_{\min}$=10 regions. The main manuscript's structure-null analysis instead works from $l_{\min}$=3 sets, so the overlap was recomputed on those (`R/12`, `R/13`, `R/15`). The three sets differ enormously in how much genome they claim:

Set	$l_{\min}$	Regions	Span	Global-spec. hit	Fold	p
EMMAX	3	17	**1.21 Mb**	**5/17**	**19.5×**	**0.0005**
LFMM	10	40	21.13 Mb	4/40	1.38×	0.33
LFMM	3	136	37.27 Mb	9/136	1.87×	0.036

The EMMAX $l_{\min}$=3 set is the standout: 17 regions covering 1.21 Mb — 0.3% of the genome — of which 5 land on the 39 Global-specific EcoPeaks, **19.5× over a within-chromosome rotation null**. Against the Pacific-specific set it is 5/17, 2.36×, $p = 0.039$; against the sensitive sets 7/17 (4.61×) and 8/17 (1.48×, n.s.). The LFMM $l_{\min}$=3 set covers 30× more genome for a 1.87× enrichment, so the difference is precision, not coverage.

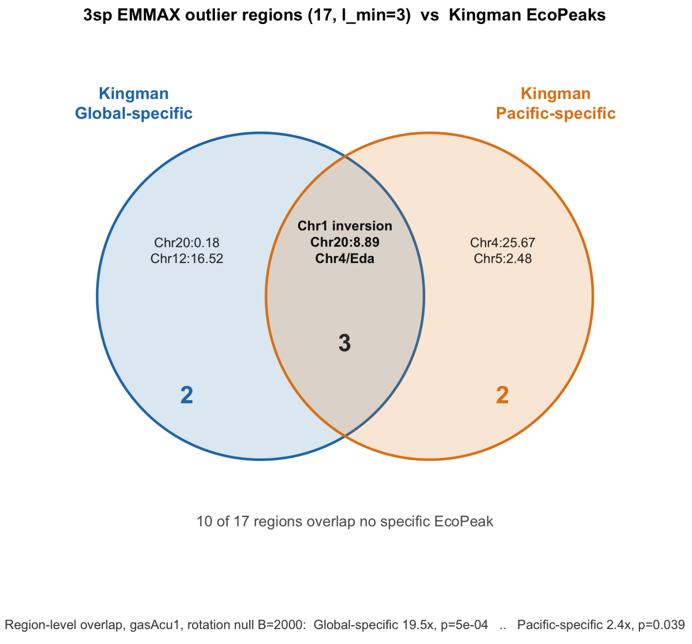

Figure 5: Which of the 17 EMMAX $l_{\min}$=3 regions hit which specific EcoPeak set. Three regions hit both — the Chr1 inversion, Chr20:8.89 Mb, and Chr4/*Eda* (§8.2). Ten of 17 hit no specific EcoPeak.

Rank-based enrichment of low Kingman p inside the 17 regions, and the c151 suggestive-peak overlap, both strengthen accordingly:

c151 suggestive peaks inside the 17 regions: 5 of 41 against 0.33 expected, **15.1×,** $p = 0.0002$.

Two cautions on reading these numbers. They are *not* comparable to the $l_{\min}$=10 folds in §7: fold enrichment scales inversely with how much genome a region set claims, and 1.21 Mb

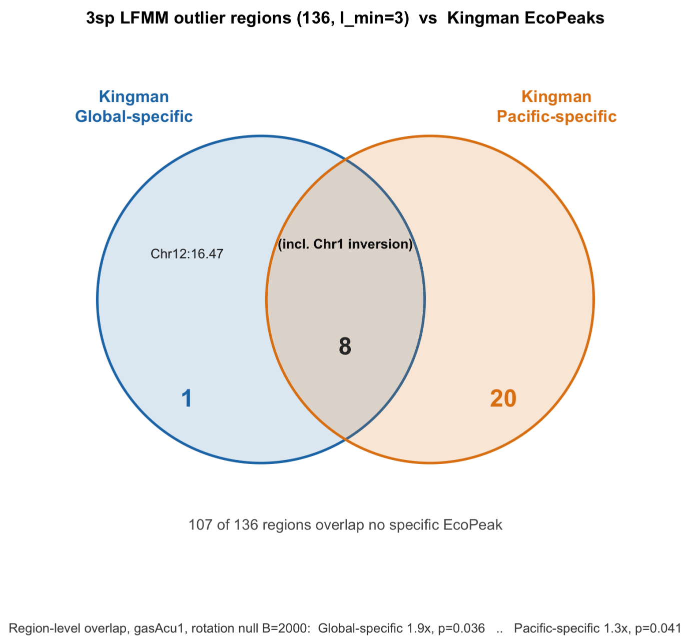

Figure 6: The same view for the 136 LFMM $l_{\min}$=3 regions, for contrast. Only 29 of 136 hit any specific EcoPeak and 107 hit none; of the nine touching the Global-specific set, eight also touch the Pacific-specific set. Note the asymmetry against Fig. 5: the LFMM set finds *more* Pacific-only regions (20 vs 2) but adds almost nothing Global-only (1 vs 2), while spending 37.27 Mb to do it against the EMMAX set's 1.21 Mb.

Kingman cohort	top 1%	p	top 0.1%	p
c151 N. Europe	**34.1**×	0.0020	11.3×	0.026
c155 Global	25.9×	0.0040	**33.5**×	0.0060
c150 N.E. Pacific	17.6×	0.0020	11.1×	0.016

of tightly targeted regions will always out-fold 19.92 Mb of broad ones. 34× here versus 4.64× there does not mean the EMMAX set agrees seven times better. And the **geographic gradient reproduces only in the top-1% column** (c151 > c155 > c150); in the top-0.1% column c155 is highest, exactly as it was for the $l_{\min}$=10 set. That is consistent across both region sets and has a straightforward reading: c155 is much the best-powered cohort (84 samples, 28 marine) and so produces the most extreme p-values, meaning the extreme tail ranks statistical power rather than geography. The top-1% comparison is where geography shows.

10 How much agreement is even achievable?

A partial overlap only means something against a ceiling. The natural ceiling is Kingman's own two published peak sets: if Global-specific and Pacific-specific — same laboratory, same VCF, same caller, differing only in which populations enter the cohort — agree only loosely with each other, then loose agreement with a 3sp region set is the expected result rather than a shortfall.

The comparison must not be made with fold enrichment, whose ceiling differs for every pair (a set covering 0.3% of the genome can reach 64×; one covering 5.6% saturates near 4×). Nor with the Jaccard index, which is capped by the span ratio of the two sets. The fair measure is the **overlap coefficient**, bp intersection over the bp of the *smaller* set: 1.0 means the smaller claim is entirely corroborated by the larger.

Set A	Set B	Span A	Span B	∩	**Overlap coef.**	p
within a dataset						
3sp EMMAX $l_{\min}$=3	3sp LFMM $l_{\min}$=3	1.21	37.27	1.158	**0.959**	0.0005
Kingman Global-spec	Kingman Pacific-spec	3.52	24.96	3.340	**0.950**	0.0005
Kingman Global-sens	Kingman Pacific-sens	22.09	82.68	20.972	0.949	0.0005
3sp EMMAX $l_{\min}$=3	3sp LFMM $l_{\min}$=10	1.21	21.13	1.046	0.865	0.0005
across datasets						
3sp EMMAX $l_{\min}$=3	Kingman Pacific-spec	1.21	24.96	0.536	0.444	0.037
3sp EMMAX $l_{\min}$=3	Kingman Global-spec	1.21	3.52	0.497	0.411	0.0005
3sp LFMM $l_{\min}$=3	Kingman Global-spec	37.27	3.52	0.972	0.277	0.049
3sp LFMM $l_{\min}$=3	Kingman Pacific-spec	37.27	24.96	4.792	0.192	0.046
3sp LFMM $l_{\min}$=10	Kingman Pacific-spec	21.13	24.96	3.173	0.150	0.028
3sp LFMM $l_{\min}$=10	Kingman Global-spec	21.13	3.52	0.476	0.135	0.32

Kingman's two datasets are essentially nested: 95% of the Global-specific sequence lies inside the Pacific-specific set, and the same for the sensitive pair. Their internal consistency is near-total — the Global peaks are, to a very good approximation, the stringent subset of the Pacific peaks. That is a high ceiling.

Our best cross-dataset figure is **0.444**, less than half of it. So the honest reading is that the 3sp region sets are *not* reproducing Kingman at the limit of what is achievable; there is a real gap, and §7 argues it is largely geographic.

One number does match the ceiling, but it is not the one that would flatter us. Our two methods agree with *each other* at 0.959 — the EMMAX $l_{\min}$=3 regions are almost entirely nested inside the LFMM $l_{\min}$=3 regions — marginally above Kingman's own 0.950. That is a within-dataset comparison, so it is not the same kind of quantity: it says the two 3sp scans are internally consistent, not that they agree with Kingman. And the containment is over a wider span ratio (1.21 Mb inside 37.27 Mb, versus 3.52 inside 24.96), which makes nesting easier. The honest summary is that **no analysis of ours agrees with Kingman as well as Kingman's two datasets agree with each other.**

All figures come from `R/10_figures.R` and every number in this report is read from `data/overlap_summary.csv`, `data/enrichment_summary.csv`, `data/c151_peak_overlap.csv` and `data/overlap_detail.csv`, all regenerated by `R/09_overlap.R` and `R/11_c151_peak_overlap.R`. The $l_{\min}$=3 results of §9 come from `data/overlap_summary_emmax17.csv`, `data/overlap_summary_lfmm.csv`, `data/enrichment_emmax1` and `data/c151_peak_overlap_emmax17.csv` (`R/12`, `R/13`, `R/15`), with Figs. 5–6 from `R/14_venn_emmax17.R` (run as `emmax17` and `lfmm`). §10 is `data/peakset_concordance.csv` from `R/16_peakset_concordance.R`.

11 Using the EcoPeaks to calibrate, not just to score

Everything above uses the EcoPeaks to *score* a fixed region set. They can also be used to *calibrate*: sweep the $\tau_C \times \ell_{\min}$ grid that produces the 3sp regions, and ask which cell best recovers an external truth. Two findings came out of doing that, one methodological and one about the two scan engines.

11.1 Most of the grid cannot support the statistic

Two of the four natural scores — precision (fraction of a cell's regions that hit an EcoPeak) and fold over a rotation null — divide by the number of regions in the cell. When a cell holds one region that happens to hit, precision is 1.0 and the fold is whatever the near-empty null allows, which on this grid reached 200×. Left unmasked these degenerate cells form a bright band across the high-τ_C, high-$\ell_{\min}$ corner and invert the apparent optimum.

Masking cells with fewer than five regions is therefore not cosmetic. It is severe for EMMAX — **only** 33 **of** 270 **cells survive** — and mild for LFMM, where 247 survive. That contrast is itself the most informative number in the sweep, and §11.2 returns to it.

Sweep	Estimable	F1-optimal cell	F_1 (max / at τ_C=0.05, $\ell_{\min}$=3)	Rank
EMMAX / global-spec	33/270	τ_C=0.10, $\ell_{\min}$=1	0.197 / 0.189	3
EMMAX / Pacific-spec	33/270	τ_C=0.05, $\ell_{\min}$=1	0.092 / 0.048	2
LFMM / global-spec	247/270	τ_C=0.10, $\ell_{\min}$=6	0.135 / 0.102	15
LFMM / Pacific-spec	247/270	τ_C=0.05, $\ell_{\min}$=5	0.217 / 0.212	4

The operating point $\tau_C = 0.05$, $\ell_{\min} = 3$ is *not* the F1 maximum in any of the four, though it is close in three. On the masked grid precision does peak in the low-τ_C corner (0.40 at $\tau_C = 0.05$, $\ell_{\min} = 6$–9), but fold does not — it favours the $\ell_{\min} = 1$ edge at high τ_C (46× at $\tau_C = 0.70$), in cells sitting at the masking floor. Recall is capped near 0.28 by biology rather than by the threshold, since most global EcoPeaks are Pacific-derived and absent from an Atlantic panel, so F1 — half of which is recall — should not be treated as the objective. Precision is the axis on which the chosen point is close to the best available.

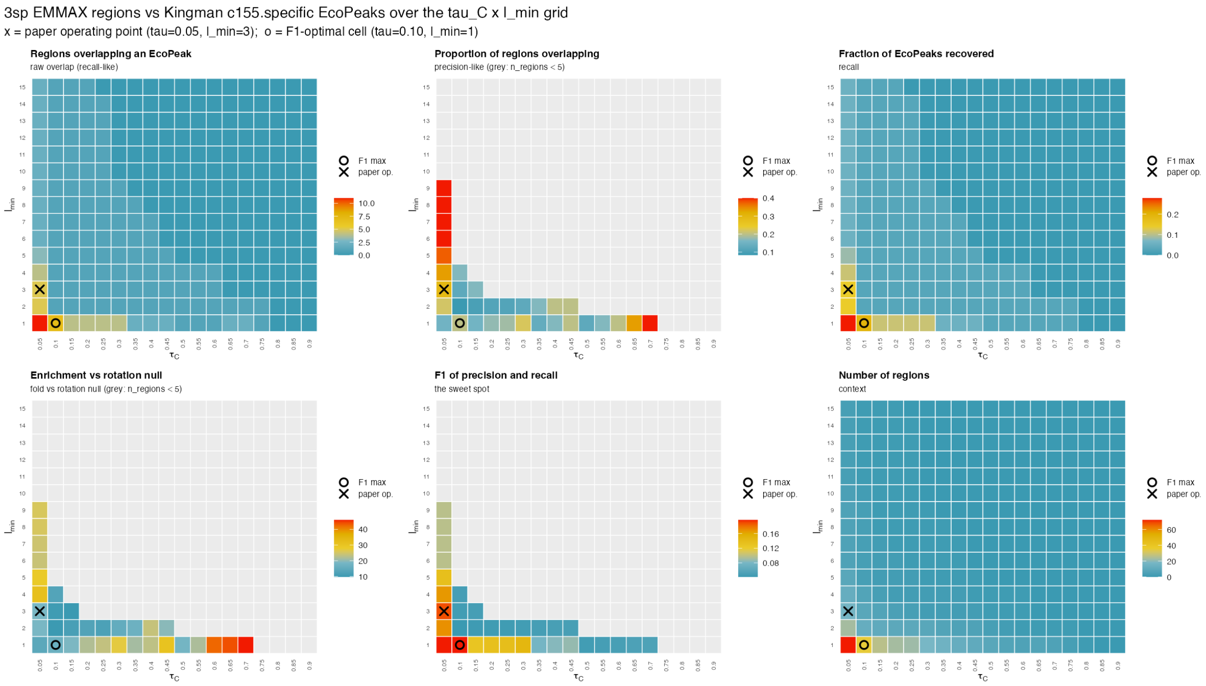

Figure 7: The $\tau_C \times \ell_{\min}$ sweep scored against the Global-specific EcoPeaks. Grey cells hold fewer than five regions, where precision, fold and F1 are not estimable. The cross is the operating point, the circle the F1 maximum.

11.2 What the sweep says about the two engines

The estimable-cell counts (33 for EMMAX, 247 for LFMM) say that C discriminates very differently in the two scans. Following $\ell_{\min} = 3$ across the grid:

Regions at τ_C =	0.05	0.10	0.30	0.60	0.90
EMMAX	17	8	2	1	0
LFMM	136	133	107	84	29

EMMAX falls out of the estimable range almost immediately; LFMM still holds 84 regions at $\tau_C = 0.60$. This matches the per-SNP picture: 782 **LFMM SNPs carry** $C > 0.5$ **against EMMAX's** 13. A large block of LFMM SNPs passes in nearly every (ρ, q^*) cell, so almost any threshold retains them.

Crucially, no threshold rescues them. Across every estimable τ_C at $\ell_{\min} = 3$, LFMM's precision stays in 0.029–0.068 and its fold in 1.8–6.4, against EMMAX's 0.125–0.294 and 9.7–18.8 over its three estimable cells. The LFMM regions are not the right regions at the wrong threshold; the ranking does not track the external truth anywhere on the grid.

11.3 The per-SNP C-score, and what it is not

Binning every SNP by C and measuring Global-specific EcoPeak membership (Fig. 8), the informative signal is the *step* from $C = 0$ to $C > 0$. EMMAX SNPs with any $C > 0$ are enriched 24–53-fold over the 1.31% genome-wide rate, while the ∼789,000 SNPs at $C = 0$ sit at background (0.91×). Within $C > 0$, however, *neither* method rises: EMMAX peaks in the $(0.05, 0.1]$ bin and falls to 23.5× at the top, LFMM falls monotonically from 21.7× to 3.8×. The positive rank correlations (Spearman $\rho = 0.21$ EMMAX, 0.06 LFMM) restate the step and are not evidence of a graded trend, since the $C = 0$ class dominates both. C should be read as a candidate flag, not as a finely graded strength of evidence.

What separates the engines is the top of the scale: at $C > 0.5$ EMMAX places 13 SNPs still enriched 23.5×, LFMM places 782 enriched 3.8×.

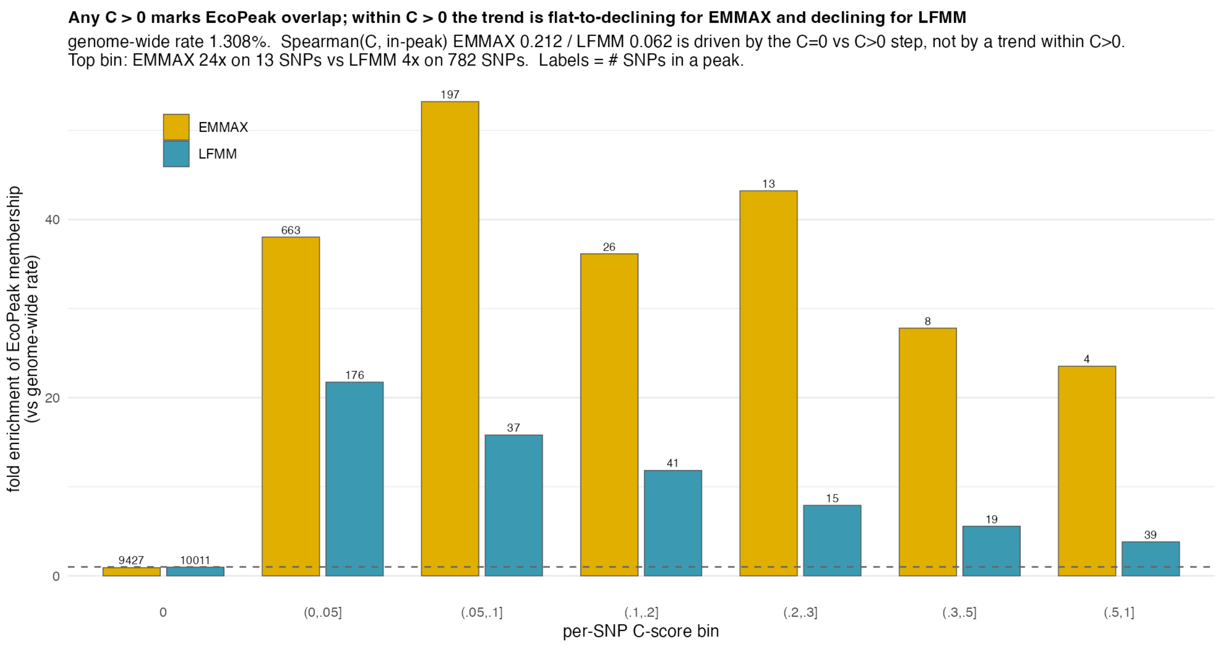

Figure 8: Per-SNP C-score versus Global-specific EcoPeak membership.

11.4 A caution: LFMM has no structure-aware null

The asymmetry above should be read with one confound in view. The EMMAX arm is calibrated by a structure-aware null; the LFMM arm has none, and its τ_C is transferred from EMMAX. Attributing the whole difference to the engines is therefore not clean.

Two observations bear on it. First, LFMM is *not* inflated in the usual sense: $\lambda_{GC} = 1.008$ against EMMAX's 1.093, because the LFMM test applies genomic control and forces the median to calibration. What it has is a heavy *tail* — 207 SNPs at $p < 10^{-6}$ against EMMAX's 13, and 952 at BH $q < 0.05$ against 3. Describing this as an inflated baseline would be wrong; the baseline is calibrated and the tail is not.

Second, the sweep and a null answer different questions. The sweep shows LFMM's C does not *rank* SNPs in a way that tracks the EcoPeaks. A null would show whether LFMM's C values are unusual *at all* against permuted ecotype. These can dissociate, and the interesting outcome is that they do: LFMM could pass its negative control — every region far above null — while failing the positive control. That would mean LFMM detects something real and structured that is not recurrent marine–freshwater adaptation.

The direction is not predictable from what is in hand, because an among-population ecotype permutation breaks the ecotype–population correlation, and if LFMM's heavy tail derives from that correlation the null will be low and its regions will clear it comfortably. This is the main open item left by this module. A small run ($B \approx$ 5–10 surrogates) would settle the direction, since what matters first is whether the null C distribution piles up near 1 or collapses toward 0; a full B is only needed for per-region p-values.

12 Are the uncorroborated regions still real?

Absence from the EcoPeaks is not by itself falsification. The EcoPeaks are an incomplete truth — one paper's cohorts, one-genome-per-population panels, and only the SNP-based half of their caller reproduced here (§11) — so a 3sp region that hits none of them might be a false positive, or might be a real Atlantic locus their cohorts cannot see. That second reading is the one that matters, because it is what would explain the LFMM surplus: 107 of its 136 regions overlap no specific EcoPeak, and if those are genuine panel-specific loci then LFMM is simply the more sensitive scan.

That reading makes a checkable prediction. If the uncorroborated regions are real marine–freshwater loci in a north-European Atlantic panel, then a north-European Atlantic cohort should see them. `c151` is exactly that (§11.4: 9 marine, 18 freshwater, Iceland to Germany), and it was *not* used to call the published EcoPeaks — only `c150` and `c155` were.

12.1 Design, and an unavoidable circularity

Each engine's $\ell_{\min} = 3$ regions were split into **corroborated** (overlapping a specific EcoPeak) and **novel** (overlapping none), and each group tested for enrichment of low `c151` p-values against a within-chromosome rotation null. The corroborated group is the **internal positive control**: `c151` cannot reach genome-wide significance on its own (§11.4), so if the corroborated regions were also flat the test would be uninformative rather than negative.

No cohort in this VCF is independent of `c155` — all 27 `c151` samples are a subset of its 84 — so the split is circular whichever way it is cut, in opposite directions. Classifying by both peak sets is *deflationary* for the novel class: a locus where the `c151` samples carry signal is likelier to have been called a `c155` peak and so labelled corroborated. Classifying by `c150` alone (which shares no samples with `c151`) is *inflationary*: `c155`-only regions fall into "novel" carrying circular signal. Both are therefore reported.

12.2 What it shows

The positive control works. Both engines' corroborated regions light up strongly and significantly (23.6× EMMAX, 4.33× LFMM), so `c151` has the power for this design despite having no FDR-significant SNP of its own. A flat result here is a real negative.

Classified by	Group	n	Span	rate	null	Fold	p
c155 + c150 *(defl.)*	EMMAX corroborated	7	0.87 Mb	0.518	0.022	23.6×	0.002
	EMMAX **novel**	10	0.33 Mb	0.0117	0.0058	2.02×	0.072
	LFMM corroborated	27	14.72 Mb	0.077	0.018	4.33×	0.016
	LFMM **novel**	107	20.98 Mb	0.0100	0.0102	**0.98**×	0.27
c150 only *(infl.)*	EMMAX **novel**	11	0.41 Mb	0.395	0.0063	62.5×	0.002
	LFMM **novel**	107	20.98 Mb	0.0100	0.0102	**0.98**×	0.27

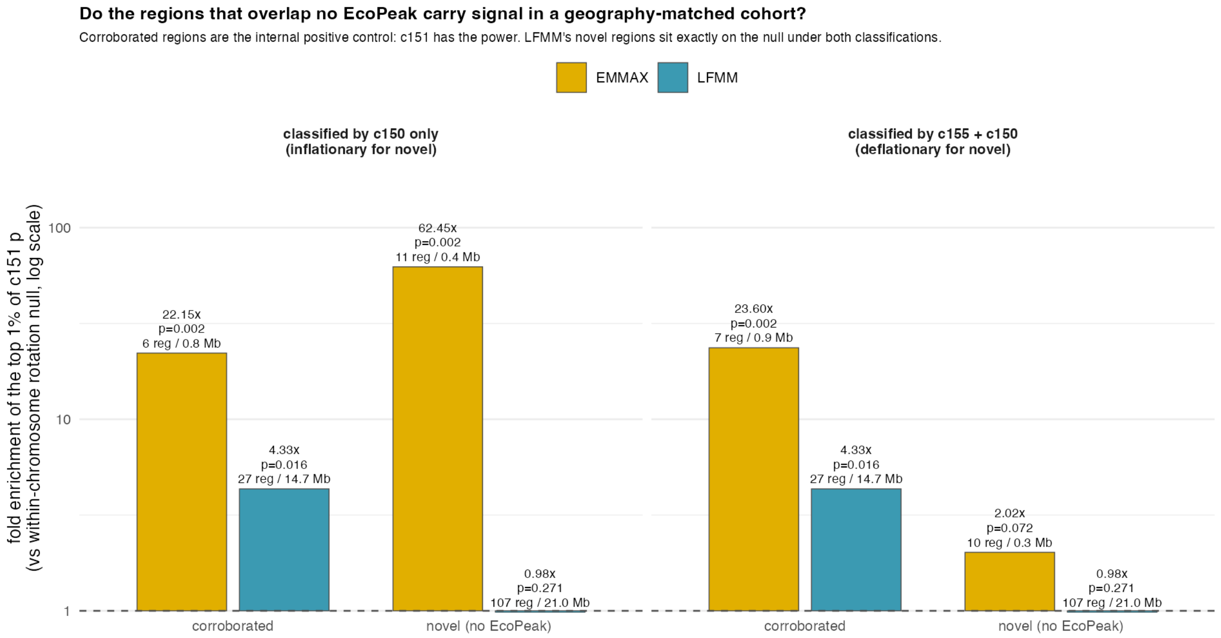

Figure 9: c151 enrichment by corroboration status, under both classifications.

LFMM's 107 novel regions sit exactly on the null: 0.98×, $p = 0.27$**, identical under both classifications.** That invariance is what makes it usable — it cannot be an artefact of either circularity direction. With 85,951 SNPs the estimate is precise rather than noisy (rate 0.0100 against an expectation of 0.0102). Nearly 21 Mb of putatively novel Atlantic marine–freshwater sequence, tested in the one cohort matched to the panel that produced it, shows nothing.

EMMAX's novel regions do carry something, but the 62.5× should not be quoted: it comes from the inflationary split, and a single region crossing the boundary moves the estimate from 2.02× to 62.5× on 11 regions. The defensible figure is the deflationary one, 2.02× at $p = 0.072$ — a positive hint, not significance, on 0.33 Mb, i.e. on 64× less sequence than LFMM's null result covers.

12.3 What it does not show

Two limits. First, zero `c151` signal is consistent with the novel regions being false positives, but equally with their being *real structure that is not marine–freshwater* — ecotype-correlated demography would look the same. This test cannot separate those, so "the LFMM surplus is false positives" is not established here; only "the surplus carries no detectable marine–freshwater signal" is. Second, it says nothing about whether those regions clear their own structure null, which is a separate question (§11.4) still awaiting the full permutation run.

It does, however, close the specific escape route that the incompleteness of the EcoPeaks opens. Panel-specific Atlantic loci are precisely what `c151` is positioned to detect, and the same argument was the remaining explanation for the declining per-SNP C-score gradient (§11).

A reconciliation with the simulations. On coalescent simulations LFMM buys recall at a modest precision cost (recall 0.325 vs 0.231; precision 0.291–0.337 vs 0.351–0.359 depending on the grid) — a genuine sensitivity advantage. On the real genome the extra regions carry no signal at all. A precision cost of order 20% in simulation does not predict a total absence, and *the gap between the two regimes is itself the evidence* for the structure-model dependence: a handful of latent factors captures the simulations' simple structure and does not capture continental-scale isolation-by-distance in the real panel.

13 Caveats

- Only the SNP-based half of Kingman's two-method peak caller is implemented; the derived sets are sensitive supersets of "specific".
- `GTs` is mean-imputed at 5.5× coverage; expect r^2/`ld_w` attenuation.
- 3sp region boundaries are the min/max SNP position of an LD cluster, hence coarse.
- 3 of 39 global-specific peaks lift to `chrUn` and 1 sits on `chrXIX`, which 3sp drops, leaving 36 comparable.
- c151 results are rank-based and carry no FDR guarantee.
- The sweep of §11 masks cells below five regions; the floor is a pragmatic choice, not a derived threshold.
- LFMM has no structure-aware null here, so the engine comparison is confounded (§11.4).
- No cohort in this VCF is independent of `c155`, so the corroborated/novel split of §12 is circular in one direction or the other; only results invariant to the choice should be relied on.
- Fold enrichments are not comparable between region sets of different total span (1.21 Mb for EMMAX $l_{\min}$=3 vs 21.13 Mb for LFMM $l_{\min}$=10); compare p-values and hit counts, not folds.
- $l_{\min}$ is consequential, not cosmetic: *Eda* is a 4-SNP cluster and is discarded by any $l_{\min}\geq 10$ analysis (§8.2).
- The recombination map leaves 3.6% of SNPs without a `rec_rate`.